

From Experience to Experience: Key Insights for Improved Interaction with AI in Radiology

Ning Fang
n.fang@tue.nl
Eindhoven University of Technology
Eindhoven, Netherlands

Jon Pluyter
jon.pluyter@philips.com
Philips Experience Design
Eindhoven, Netherlands

Saskia Bakker
sbakker@cinop.nl
Philips Experience Design
Eindhoven, Netherlands

Igor Jacobs
igorjacobs85@gmail.com
Philips Research
Eindhoven, Netherlands

Misha Luyer
misha.luyer@catharinaziekenhuis.nl
Catharina Ziekenhuis
Eindhoven, Netherlands

Joost Nederend
joost.nederend@catharinaziekenhuis.nl
Catharina Ziekenhuis
Eindhoven, Netherlands

Jeroen Raijmakers
J.H.M.Raijmakers@tudelft.nl
Philips Experience Design
Eindhoven, Netherlands

Lin-Lin Chen
l.chen@tue.nl
Eindhoven University of Technology
Eindhoven, Netherlands

Mathias Funk
m.funk@tue.nl
Eindhoven University of Technology
Eindhoven, Netherlands

ABSTRACT

Artificial Intelligence (AI) decision-making tools for radiology demonstrated potential capacity to improve radiology work in several tasks such as tumor detection. However, relatively low acceptance in clinical practice demonstrates the challenge of incorporating end-users' lived experience and their opinions to improve the interaction between clinicians and AI solutions. Therefore, we conducted semi-structured interviews with radiologists and technicians who had lived experience with current or the prior generations of radiology AI tools (e.g., Computer Aided Decision tools). Three key themes were elicited. Firstly, *the role of AI*, addresses how radiology professionals interact with radiology AI; the second theme, *adoption in practice*, discusses the requirements for easy usage and smooth transition; the third theme, *building appropriate trust*, explores influencing factors of clinicians' trust towards radiology AI. Our findings call attention to the adoption of actionable recommendations on the interaction design and the importance of individual tailored functionalities in radiology AI systems.

CCS CONCEPTS

• **Human-centered computing** → *Interaction design theory, concepts and paradigms*; • **Applied computing** → **Life and medical sciences**.

KEYWORDS

Human-AI collaboration, Radiology, Healthcare AI, Adoption

ACM Reference Format:

Ning Fang, Jon Pluyter, Saskia Bakker, Igor Jacobs, Misha Luyer, Joost Nederend, Jeroen Raijmakers, Lin-Lin Chen, and Mathias Funk. 2024. From

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3650779>

Experience to Experience: Key Insights for Improved Interaction with AI in Radiology. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3613905.3650779>

1 INTRODUCTION

Artificial Intelligence (AI) based decision-making tools are increasingly being recognized as promising applications in clinical radiology [20]. With the large technological improvement and the tighter collaboration between domain experts and system development teams, existing AI-based decision-making tools for radiology has shown promise in various tasks, such as decision support, operational optimization, and patient empowerment [5]. For instance, certain AI tools for radiology can diagnose pneumonia on chest X-ray images more accurately than radiologists for specific benchmarks [27]. Therefore, healthcare professionals, including clinicians and policy makers, anticipate significant gains in efficiency and quality for healthcare through the contribution of AI-based decision-making tools [20]. Although such expectation gives radiology professionals a promising vision about the way of working in the future, it creates also the “hype” about AI applications driven by simple assumptions based on the potential of technology alone [12].

1.1 Adoption of Artificial Intelligence in Clinical Radiology

As a high-stakes area where life-critical decisions are made in a short timeframe [28], increasing demands have been placed on radiology professionals to perform fast and accurate assessment. Given such unique nature of radiology work, clinicians have high working pressure and rely on strong routines [33]. Moreover, the number of images per case has increased significantly in recent years, which results in even higher workload for radiologists [8]. Furthermore, the costs of failure in daily work and the need for reliability are typically high in this specific field [36]. Consequently, as demonstrated by researchers and medical companies, the perceived potential of radiology AI positions it as a helpful partner with promise for the future of the field. Despite the potential promise of AI-based

decision-making tooling and the availability of various radiology AI products in the market, their successful adoption in clinical practice is often fraught with challenges [21, 24, 33, 36]. Only limited number of applications has been translated into routine clinical care [9]. A study collects information of applying AI products in radiology in 54 out of 69 hospitals in the Netherlands between 2020 and 2022, showing that the number of total implementations is increasing [34]. However, only 14% of general hospital respondents reported using radiology AI until 2022 [34]. On a broader European scale, a survey among 690 radiologists' members of the European Society of Radiology revealed that only 13.3% of them said yes when asked if they would like to use AI for their practice [26]. Regarding working with AI, 48% of the respondents anticipated an increase in their workload [26]. Accordingly, a hype-dissatisfaction cycle might appear in the society [6].

The discrepancy between the high promise of AI for radiology and its adoption reality illustrates the challenges of facilitating the acceptance and implementation in clinical settings. It also reveals that the true complexity of clinical practice is not embraced in technology development. Clearly, high performance alone is not enough for AI to enter daily work of clinicians, though technical concerns do play an important role here [30, 32]. One significant barrier to successful implementation is the lack of understanding of the interaction between end-users and AI solutions [21]. Adopting such a novel technique means behavior changes of clinicians, including adjustments to established routines [12]. Changing existing habits is tough in general, which brings the need to design clinician-AI interactions that can eliminate the friction of adjusting to the new system as much as possible since these AI systems are made to support clinical routines. In this regard, more clinician-AI interaction focused research and tests in practices are needed [14]. To realize this objective, it is crucial that end-users' voices are present in the AI system development process.

1.2 End-user Experience and Attitudes

Prior research has focused on understanding the attitudes of radiology professionals regarding AI-based decision-making tool implementation [11, 17, 25, 33, 34]. Radiologists showed overall positive attitudes towards the integration of AI systems into their daily work routines. An overarching belief is held that AI for radiology could optimize workflows [17] and automate routine manual tasks in the future [25]. For example, in terms of the role of radiology AI, it is most frequently conceptualized as a *second reader* instead of a replacement of radiology professionals [17]. In contrast, clinicians also expressed their concerns about this trending technique. Many of them worry about *inconsistent technical performance* of the AI-based systems, which may undermine the establishment of appropriate trust in their reliability [33, 34]. Finally, factors pertaining to hospital policies will also influence the use of radiology AI, including uncertainties in funding, regulations, and legality [33]. There are knowledge and experience gaps between radiologists and designers, data scientists, and developers of AI-based solutions [23]. The lack of knowledge such as anatomical concepts and correspondingly the working experience and operational pain points [23] or the black-box nature of deep learning models make it difficult for radiologists to collaborate in algorithm design and

Table 1: Overview of participants

	Role	Gender	Working experience
P1	Radiologist	F	>20 years' experience
P2	Radiologist	M	>20 years' experience
P3	Radiologist	M	In training
P4	Radiologist	M	In training
P5	Radiologist	F	5-10 years' experience
P6	Lead technician	F	>20 years' experience
P7	Technician	M	10-20 years experience
P8	Radiologist	M	0-5 years' experience

assessment [30]. Moreover, most data was collected via large scale surveys or interviews among clinicians without experience of using AI [1, 17, 37], leaving the lived experience with radiology AI inadequately presented [16].

In this paper, we focus on 1) delivering specific experience from end-users who had experience with earlier generations of radiology AI and 2) providing their recommendations for the design of radiology AI. Our study investigates layers of attitudes of end-users towards AI-based decision-making tools for radiology, and what matters to them in their interaction with radiology AI. In the following, we first explain methodology, showcase the results, discuss the emerging patterns, and conclude with an outlook on future work.

2 METHODOLOGY

To cover AI use across the radiology imaging workflow, we included 6 radiologists and 2 senior radiology technicians from a hospital in the Netherlands (Table 1). Technicians are responsible for acquiring the appropriate images at the scanner, whereas radiologists analyze and report their findings to the referring physician. All participants actively work with AI-based tools or CAD in their practice, such as CVI24 for cardiovascular imaging and lung emboly detection software. Four of our participants have more than 10 years of working experience, two of them have 0-10 years' experience, and two are in-training radiologists.

We conducted semi-structured interviews with each participant for 30-45 minutes for each session. Two technicians were interviewed as a pair to minimize daily work disruptions and are referred to as P6P7 in this article. We asked them open-ended questions about 1) their experience with actual use of AI in clinical settings; 2) they key enablers and roadblocks they experienced or foresee, and 3) their vision of an ideal future of using AI. Clinicians participated voluntarily after providing informed consent, no financial compensation was provided.

All interviews were done in the local language and later transcribed to English by native speakers. We developed findings using reflexive thematic analysis according to the six-phase guide identified by Braun and Clarke [7]. The data analysis was performed using NVivo. After reading through the transcripts to familiarize with the data and noting down the initial patterns, we generated codes from personal experience and related literature. Then the codes were selected and sorted into initial themes. After that, we reviewed and refined the themes in an iterative process.

3 RESULTS

We found three main themes to capture what matters to clinicians when they use radiology AI in daily practice based on their understanding and previous experience about it (Table 2). The first theme, *designing the role of AI*, includes where and how participants think AI will play a role in their workplace and how they can collaborate; the second theme, *adopting AI in practice*, discusses the requirements pertain to workflow for radiology AI to be commonly used in clinical practice; the third theme, *building appropriate trust, conceptually and in situ*, explores the factors that facilitate appropriate trust of clinicians towards AI for radiology.

3.1 Designing the Role of AI: Radiologist-AI Interaction

Most participants discussed what value they foresee that AI can bring to radiology to benefit patients, clinicians, and healthcare systems. One participant addressed that the overarching objective of technological innovation is the improvement of patients' health: *"Eventually don't forget that it is about what is important for patients"* [P5]. In terms of the collaboration between clinicians and AI, participants pointed out some domains they care about the most in the practice and might be the roadblocks of implementing AI systems as well.

3.1.1 AI as "safety net" or concurrent reader? To enhance efficiency and support clinicians in their work, designing the moment when AI play a role is crucial. It was discussed a lot whether AI suggestions should serve as a first or a second reader. More than half of the participants reported that they use AI as a "safety net", which can be understood as a reference that clinicians can check with after their initial scan readings and idea formulation. Our participant gave an example of how AI is used as a safety net: *"When I was done with reading the case myself, I pressed the button to see whether the computer found polyps that I missed. I use it at the moment as a 'safety net'"* [P1]. *"Then you talk about the interface? Well, the program starts such that you can first look at the case yourself"* [P8].

In this case, being as safety net, AI functionalities can be activated with buttons next to the scans to ensure clinicians do not miss findings. Such approach also preserves clinicians' independent thinking process and autonomy. Specifically, AI can be a compensation to the "pitfalls" of radiologists: *"That is where I need support. Because you know your pitfalls, and even though you pay special attention to it, it happens over and over again"* [P1]. Participants highlighted that they see AI also as a confirmation to their preliminary decisions: *"If the software does not indicate lung embolies, then the probability that there is a lung emboly is very low, so if I do not see an emboly myself either then you are sure there are no lung embolies"* [P4].

Regarding the role of AI as a concurrent reader, in contrast to being a second reader, participants spoke about the advantages of having instant answers alongside the scan but also the concern about being biased. On the one hand, for some tasks like checking whether there is blood in the skull of trauma patients, AI can effectively filter out the negative cases and ease the work for radiologists. *"If you would have a CADx that screens for you, as a sort of filter, that would be fantastic for certain disease areas"* [P2]. On the other

hand, people do not trust AI fully and are afraid of being biased or getting a lot of false positives that could lead to unnecessary scans and consequently extra costs for patients. *"So what you catch might turn out to be nothing or accidental findings that you need to burden someone, where additional diagnostics are needed and the costs grow out of hand"* [P5].

3.1.2 Communicating capabilities truthfully. The information provided by AI can influence clinicians' decision-making processes to some degree. Thus, in addition to when AI jump into the party, the content displayed in the system also matters. In addition to conclusive outcomes such as the benign or malignant nature of a tumor, participants expressed the desire for more detailed information to aid their decision-making. One important information is the strengths and the weaknesses of the model. After learning the strengths and the weaknesses of the systems, clinicians can be more proficient [13]. Besides, similar to discussing patient cases with a colleague, awareness of the capabilities of an AI model enables radiologists to formulate appropriate actions: *"So that you know what you can do with it, what you cannot do with it, what the dangers are"* [P2].

Another information that radiologists would like to see along with the conclusion is the probability of it. With the percentage saying how sure AI is about the conclusion, *"you get a number which you can interpret yourself"* [P3]. Instead of providing merely the results, AI assists decision-making by presenting similar "thinking" process as clinicians, that is, showing the level of its confidence. This can be considered by clinicians just like discussing cases with their colleagues.

3.2 Adopting AI in Practice

Accepting a new technology is usually a long and tough process, particularly during its initial stages. The introduction and continuous use of AI in clinical practice may encounter certain frictions result from factors such as high workload environment. Such phenomenon suggests that implementing an adaptive system and ensuring a good onboarding experience are necessary.

3.2.1 Easy, available, applicable. Adopting, or even embracing AI for clinicians can be a big step if they need to make substantial effort into applying its functionalities. Many participants reported the preference for a system that demands minimal additional steps and actions. Therefore, it is essential to make AI accessible at the fingertips. A participant summarized it: *"It should be easy to use, easy to find, easy to apply"* [P1].

"Easy to find" means AI functionalities are *"as easily available as possible"* [P3]. Thus, users do not need to search for the button in a hidden list in the software to activate the function. In terms of simple usage, participants emphasized the low threshold of using the system if only minimum steps are required. For example, AI can be switched on and the results are shown with one or two clicks. Furthermore, it is important to simplify the process of applying AI models to save time and avoid over-loading work for clinicians. *"We can register images in certain radiology AI software, that can even be done automatically, but still hardly anybody uses it because it is laborious to get those images there and apply it"* [P3]. *"For sure not that you have to open a separate application, load the images, load*

Table 2: Themes, subthemes, and example quotes

Themes	Subthemes	Examples
Designing the role of AI: radiologist-AI interaction	AI as “safety net” or concurrent reader?	“When I was done with reading the case myself, I pressed the button to see whether the computer found polyps that I missed. I use it at the moment as a ‘safety net.’” [P1]
	Communicating capabilities truthfully	“I think you want to know what you’re up to. If you get your results back, then you learn what the algorithm is good at.” [P3]
Adopting AI in practice	Easy, available, applicable	“That you say, I just want to know what AI thinks, and that it is there, on and off. It needs to be that simple in my view.” [P8]
	Incumbent workflows and legacy systems	“It would be good if that is integrated [in the PACS] and that if you click the findings, you see them direction on the CT.” [P8]
	Less onboarding, more learning and anticipating	“There is an implementation phase, and everybody needs to learn to work with it.” [P4]
Building appropriate trust, conceptually and in situ	Robust but more so predictable performance	“I think the trust will only be there, once AI sees something of which you are not sure and you do additional exams that confirm the AI finding, then I think trust will grow.” [P8]
	Showing how AI gets to a result	“That could be useful to see how AI got to the conclusion, if for example it would interpret the abdominal mass, maybe AI goes into a certain ‘decision tree’ that you can follow. Then you can see, it is going down this path but it saw this, so I would have gone a different path.” [P6P7]
	Domain knowledge is personal and experiential	“We have a lot of knowledge to know how to go about with data. I think it is important that radiologists are involved, and that algorithms are put in the market through us. Not via all sorts of companies. Some companies are big players, but I find it important that we are involved.” [P2]

the patient data. So actually, if there are multiple steps in between, then people don’t use it” [P3].

3.2.2 Incumbent workflows and legacy systems. To comprehend the importance of seamlessly integrating the new system into the current workflow of radiologists, it is essential have an idea about their complex software workstation. Currently many radiologists are working with two medical systems, an image viewer (e.g., Picture Archiving and Communication System (PACS)) and a patient record system connected to the viewer.

To achieve workflow integration, software integration is necessary [4]. Radiologists expressed that the threshold of using AI would be higher if they need to open a separate software to run it. Moreover, they are already used to the existing systems that are satisfying, it would be a burden to adapt to working with AI and a new software alongside their demanding schedules. *“It would be good if that is integrated in the PACS and that if you click the findings, you see them directly on the CT” [P8].*

3.2.3 Less onboarding, more learning and anticipating. When asked about potential roadblocks of integrating AI into daily clinical practice, one participant highlighted the challenge of the learning process, “I think also the learning process towards it” [p4]. Radiologists and technicians work in a high-stakes and high-pace environment, it is less possible for them to dedicate significant time on learning and understanding how to use and work with AI. However,

given that onboarding can play an important role in adopting a new technology, designing an efficient and positive experience for our end-users is a must.

3.3 Building Appropriate Trust, Conceptually and in situ

Besides the value that AI can bring to clinicians and practical requirements for its integration into daily workflow, trust has been under the spotlight because it is mentioned as a prerequisite for harnessing clinical-AI in the practice and both under- and over-trust can impede the adoption [19]. Our participants also addressed trust as an essential element throughout the application process. As trust is a multifaceted construct, clinicians discussed how trust towards AI can be built up from different perspectives.

3.3.1 Robust but more so predictable performance. The performance of AI system is a focal topic when it comes to the real-life application, and it is essential for fostering appropriate trust. Participants expressed that AI becomes convincing to them when *“in a certain subset of exams you see that AI outperforms the radiologist” [P5]*. Moreover, from the perspective of our participants, accuracy weighs a lot in indicating the performance of AI. *“Only when you see that often it goes right, then you build trust, and I think you cannot build that in a different way” [P3]. “I think the trust will only be there, once AI sees something of which you are not sure and you*

do additional exams that confirm the AI finding, then I think trust will grow” [P8]. Besides the performance at the moment, the ability to demonstrate the potential for improved accuracy over time is equally important in convincing clinicians and gaining their trust: “If you can show that you have for example a 30% better performance than before, then you gain trust. Those statistics you should be able to see yourself when you want it. For example, every month, or every three months. Do I see the trend going up or down?” [P6P7]

3.3.2 Showing how AI gets to a result. Different from providing results and the probabilities, showing how the conclusions are generated can help with the comprehension of the inner workings of AI, which brings clinicians closer to how AI “thinks”. *“That could be useful to see how AI got to the conclusion, if for example it would interpret the abdominal mass, maybe AI goes into a certain ‘decision tree’ that you can follow. Then you can see, it is going down this path but it saw this, so I would have gone a different path” [P8].*

However, some participants spoke about not being interested in knowing the rationales behind the model unless *“it is important to know parts of it because thereby I can control the algorithm better than I would like to know” [P1].* This indicates that the level of detail in explaining how AI operates should align with the importance of the information for decision-making for end-users.

3.3.3 Domain knowledge is personal and experiential. When we consider infusing end-user perspectives in the design process, the importance of their previous experience and lived experiences matters as early as the conceptualization stage. Participants emphasize their desire to actively participate in the development process, contributing to shaping the system by providing experience from the field. *“We have a lot of knowledge to know how to do with data. I think it is important that radiologists are involved, and that algorithms are put in the market through us, not via all sorts of companies.” [p2].* In addition to shaping ideas, our participants expressed the need to experience the radiology AI systems before it is ready to be implemented in the field, so that they have an idea about the capacity of radiology AI tools. As AI applications are rather new for many radiologists, it is easier for them to evaluate the system and even build a collaboration scheme based on their working habits after experiencing it: *“I think I should experience that in order to be able to say if that would be useful, I find it difficult to imagine” [P1].*

Overall, participants shared their ideas about where and how they would like to have radiology AI to assist their daily work, as well as some possible hurdles. Specifically, from the way of interaction with radiology AI tools, the transition process, and clinicians’ appropriate trust towards the systems, radiology professionals provided some concrete examples and context of using such tool with the combination of their own experience with AI-based decision-making tools.

4 DISCUSSION

Designing AI solutions for *and with* radiology professionals is a vital yet challenging task, given the high stakes and time pressure nature of their daily work. We have found three themes for clinicians engaging with radiology AI tools. Firstly, the role of AI

in the interaction should be clear and better tailored to individuals. Slightly different from previous research that indicated preference of radiology AI being a second reader [1], our results reveal that radiology professionals are open to using AI-based decision-support tools in both second and concurrent reader roles according to types of exams, situations, and their own preferences. This could be partly explained by the wide coverage of functionalities and specializations in radiology AI systems, and each possessing various characteristics [5]. Such contextual customization should also be considered regarding the information presented in the system, which also underlines the importance of software customization in interface design and functional workflows. Secondly, given the time constraints in clinical environment, the transition process should be as seamless as possible. Learning and getting used to a novel decision-making tool might cause extra unwarranted time and possible resistance as also shown by Al Sharif [2]. As reported by our participants, designing user-friendly interaction behavior and integrating radiology AI with existing software can effectively alleviate the perceived “transition pain”. In addition, it has been shown that a well-designed onboarding program can facilitate user adoption [22] as it can be a testing and learning environment where clinicians are under less pressure. The third theme highlights that we should design for appropriate trust between radiology professionals and an AI-based system. Our participants reported that robust AI performance represents a direct pathway to gain appropriate trust over time. Unlike the conclusions drawn in previous research, suggesting that trust is influenced by several factors [3], our participants expressed a relatively straightforward understanding of building trust in radiology AI systems. Many of them believe that radiology AI adoption would be a natural process as the introduction of other innovations such as airplanes. Thus, appropriate trust in radiology AI would be built if it proves its performance and reliability. Besides, radiology professionals’ desire to participate in different development phases to enhance acceptance supports the needs of user-centered approach [15]. As argued by Chen et al. [10], we need to ensure that AI solutions meet user needs to facilitate the adoption of it in daily practice.

4.1 Design Implications for Radiology AI

Design implications go beyond providing insights that designers can act on more easily, we want to raise questions that need to be answered in-process by experimenting and iterating on prototypes, preferably in the field. Firstly, we need to be aware that there is no “one-size-fits-all” solution. It is important to investigate various conditions where AI can play certain roles in certain diseases for specific clinicians, i.e., whether the role of radiology AI is a second reader or a concurrent reader is a pivotal question, amongst many others. The interviews reveal that AI can play different roles according to the complexity of the task, the type of the task, and the preference of the clinician. Therefore, to optimize when to use which approach for specific clinician, designers need to consider where the *“cut-off” point will be set within the system* [P8] and possibly the degree of customization based on the preference of clinicians. Besides receiving assistance in decision-making, participants spoke about using AI can be a detour in their daily workflow.

There are established workflows or standards to guide decision-making for each disease. In this case, for each disease-specific AI solution, we need to know its conventional workflow and identify where AI can be positioned in order not to interrupt the flow. In terms of the workflow for each clinician, not only their personal working style should be considered, but their preference of using AI as well. Precisely, the workflow of dealing with one disease for one radiology can be different accordingly, therefore individual-tailored, disease-tailored, or the combination of both, and the degree of each approach should be investigated further. One way to facilitate the initial transition is to offer examples of how to use the software. One participant noted, “*I have not seen yet a specific piece of software that exemplifies how we should do it*” [P3]. Performing trial sessions in their daily work may prove helpful for them to get familiar with the system functionalities and capabilities. This hands-on experience can contribute to a smoother and successful integration of AI into clinical practices. The method and style of introducing an AI-based tool also need to be tailored across different settings. For example, the variation of overall knowledge of radiologists in the hospital and the organizational features might influence the way of using radiology AI [13]. Given that the diagnosis standards are different across diseases, pointing out the specific area contributing to the conclusion is also helpful. One radiologist gave an example pertains to liver scans: “*...if there are density differences, what shape does it have, which HU (Hounsfield unit) values (I need to look at)*” [P5]. In the same vein, for some other diseases, in addition to the diagnosis or the results, radiology AI tools can pinpoint the precise slide or area that contributes to the conclusion to help with discussion and final decision-making. The discussion also highlights the need to consider the extent to which and the way how clinicians are explained about AI rationales [18]. It is important to strike a balance to ensure the explanation is sufficient for clinicians to understand how AI works without exposing them to excessive information.

Besides, results about the view of clinicians on trust in their collaboration with radiology AI are interesting. Existing literature characterizes trust as a complex and abstract concept that is hard to understand and build [3]. For instance, various definitions of trust converge on three key elements, which are vulnerability, positive expectations, and attitude [35]. However, our interviews with radiologists and technicians suggest a relatively straightforward interpretation. Most of them reported that radiology AI systems would gain trust from them if it proved its accuracy. More importantly, the necessity of experiencing the performance to cultivate trust was highlighted. These responses indicate that trust is a multi-layer concept and individuals can easily be unaware of it, which raises the urgency and the necessity to further investigate the influencing factors of trust under awareness. It should also be noted that clinical practice has its unique characteristics. For some influencing factors of trust that has been discussed in theories or experiments in labs, they failed to prove its role in building appropriate trust in reality. For instance, explanations for model predictions were found unusable in clinical settings due to time pressure in this critical environment. Future work should focus on *appropriate trust* in radiologist-AI interaction and further investigate determinants of clinical acceptance and adoption related to trust, transparency and AI-based solution design.

Furthermore, we suggest that radiology AI tools development teams should incorporate feedback of radiology professionals in different phases to ensure that our AI-based tools can be iterated in a way that benefits end-users by tackling their needs more directly [29]. Throughout the development process, there are several roles that radiology professionals can play, in problem shaping, algorithm critique, and AI interaction test [31]. Moreover, future research can address methods for end users to provide feedback based on their experience working with AI systems as AI models are updated in fast-paced iterations [14].

5 CONCLUSION

This paper reports on an interview-based study on the interaction preferences of radiologists for AI-based decision-making tools. We surface key considerations shaping attitudes and expectations for interaction and collaboration with radiology AI. Building on existing studies about clinicians’ attitudes towards radiology AI, we identify deeper layers of attitudes from the field. Three main themes were elicited from the interviews. Participants addressed that 1) the role of AI for radiology during the interaction with professionals should be designed according to specific needs; 2) easy to use and integrating into workflows is necessary for radiology AI to be adopted in the practice, and 3) robust performance, showcase certain AI working process, and involving end-users may foster appropriate trust towards AI systems. Our findings confirm previous research results regarding the attitudes of clinicians [11, 17, 25, 33, 34]. We further extend prior findings with empirical recommendations about the direction of improvement of radiology AI. This study also enrich the understanding of the usage of the current generation of radiology AI from clinical practice. Reflecting on our findings, we provide design implications for such systems, suggesting a better connection between end-users and AI system designers, and appreciation of expert-provided experiential recommendations.

REFERENCES

- [1] Anjali Agrawal, Garvit D Khatri, Bharti Khurana, Aaron D Sodickson, Yuanyuan Liang, and David Dreizin. 2023. A survey of ASER members on artificial intelligence in emergency radiology: trends, perceptions, and expectations. *Emergency Radiology* 30, 3 (2023), 267–277. <https://doi.org/10.1007/s10140-023-02121-0>
- [2] Walaa Alsharif, Abdulaziz Qurashi, Fadi Toonsi, Ali Alanazi, Fahad Alhazmi, Osamah Abdulaal, Shrooq Aldahery, and Khalid Alshamrani. 2022. A qualitative study to explore opinions of Saudi Arabian radiologists concerning AI-based applications and their impact on the future of the radiology. *BJR| Open* 4 (2022), 20210029. <https://doi.org/10.1259/bjro.20210029>
- [3] Natalie C Benda, Laurie L Novak, Carrie Reale, and Jessica S Ancker. 2022. Trust in AI: why we should be designing for APPROPRIATE reliance. *Journal of the American Medical Informatics Association* 29, 1 (2022), 207–212. <https://doi.org/10.1093/jamia/ocab238>
- [4] Seth J Berkowitz, Jesse L Wei, and Safwan Halabi. 2018. Migrating to the modern PACS: challenges and opportunities. *RadioGraphics* 38, 6 (2018), 1761–1772. <https://doi.org/10.1148/rg.2018180161>
- [5] Tom Boeken, Jean Feydy, Augustin Lecler, Philippe Soyer, Antoine Feydy, Maxime Barat, and Loïc Duron. 2023. Artificial intelligence in diagnostic and interventional radiology: where are we now? *Diagnostic and Interventional Imaging* 104, 1 (2023), 1–5. <https://doi.org/10.1016/j.diii.2022.11.004>
- [6] Mads Borup, Nik Brown, Kornelia Konrad, and Harro Van Lente. 2006. The sociology of expectations in science and technology. *Technology analysis & strategic management* 18, 3-4 (2006), 285–298. <https://doi.org/10.1080/09537320600777002>
- [7] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative research in psychology* 3, 2 (2006), 77–101. <https://doi.org/10.1191/1478088706qp0630a>
- [8] RJM Bruls and RM Kwee. 2020. Workload for radiologists during on-call hours: dramatic increase in the past 15 years. *Insights into imaging* 11 (2020), 1–7. <https://doi.org/10.1186/s13244-020-00925-z>

- [9] Josip Car, Aziz Sheikh, Paul Wicks, and Marc S Williams. 2019. Beyond the hype of big data and artificial intelligence: building foundations for knowledge and wisdom. , 5 pages. <https://doi.org/10.1186/s12916-019-1382-x>
- [10] Haomin Chen, Catalina Gomez, Chien-Ming Huang, and Mathias Unberath. 2022. Explainable medical imaging AI needs human-centered design: guidelines and evidence from a systematic review. *NPJ digital medicine* 5, 1 (2022), 156. <https://doi.org/10.1038/s41746-022-00699-2>
- [11] Yaru Chen, Charitini Stavropoulou, Radhika Narasinkan, Adrian Baker, and Harry Scarbrough. 2021. Professionals' responses to the introduction of AI innovations in radiology and their implications for future adoption: a qualitative study. *BMC health services research* 21, 1 (2021), 1–9. <https://doi.org/10.1186/s12913-021-06861-y>
- [12] Ezekiel J Emanuel and Robert M Wachter. 2019. Artificial intelligence in health care: will the value match the hype? *Jama* 321, 23 (2019), 2281–2282. <https://doi.org/10.1001/jama.2019.4914>
- [13] Nuša Farič, Sue Hinder, Robin Williams, Rishi Ramaesh, Miguel O Bernabeu, Edwin van Beek, and Kathrin Cresswell. 2024. Early experiences of integrating an artificial intelligence-based diagnostic decision support system into radiology settings: a qualitative study. *Journal of the American Medical Informatics Association* 31, 1 (2024), 24–34. <https://doi.org/10.1093/jamia/ocad191>
- [14] Ross W Filice and Raj M Ratwani. 2020. The case for user-centered artificial intelligence in radiology. , e190095 pages. <https://doi.org/10.1148/ryai.2020190095>
- [15] Bo Gong and Michael N Patlas. 2023. Learning Radiology AI Concepts by Seeing and Doing. , 08465371231151239 pages. <https://doi.org/10.1177/08465371231151239>
- [16] David Hua, Neysa Petrina, Noel Young, Jin-Gun Cho, and Simon K Poon. 2023. Understanding the factors influencing acceptability of AI in medical imaging domains among healthcare professionals: A scoping review. *Artificial Intelligence in Medicine* 9 (2023), 102698. <https://doi.org/10.1016/j.artmed.2023.102698>
- [17] Merel Huisman, Erik Ranschaert, William Parker, Domenico Mastrodicasa, Martin Koci, Daniel Pinto de Santos, Francesca Coppola, Sergey Morozov, Marc Zins, Cedric Bohyn, et al. 2021. An international survey on AI in radiology in 1,041 radiologists and radiology residents part 1: fear of replacement, knowledge, and attitude. *European radiology* 31 (2021), 7058–7066. <https://doi.org/10.1007/s00330-021-07781-5>
- [18] Weina Jin, Jianyu Fan, Diane Gromala, Philippe Pasquier, and Ghassan Hamarneh. 2021. EUCA: The end-user-centered explainable AI framework. *arXiv preprint arXiv:2102.02437* (2021). <https://doi.org/10.48550/arXiv.2102.02437>
- [19] Wiard Jorritsma, Fokke Nossen, and Peter MA van Ooijen. 2015. Improving the radiologist–CAD interaction: designing for appropriate trust. *Clinical radiology* 70, 2 (2015), 115–122. <https://doi.org/10.1016/j.crad.2014.09.017>
- [20] Indra Joshi and Jessica Morley. 2019. Artificial intelligence: how to get it right, putting policy into practice for safe data-driven innovation in health and care. *London: NHSX* (2019). https://www.nhs.uk/media/documents/NHSX_AI_report.pdf
- [21] Christopher J Kelly, Alan Karthikesalingam, Mustafa Suleyman, Greg Corrado, and Dominic King. 2019. Key challenges for delivering clinical impact with artificial intelligence. *BMC medicine* 17 (2019), 1–9. <https://doi.org/10.1186/s12916-019-1426-2>
- [22] Matthew D Li and Brent P Little. 2023. Appropriate Reliance on Artificial Intelligence in Radiology Education. *Journal of the American College of Radiology* 20, 11 (2023), 1126–1130. <https://doi.org/10.1016/j.jacr.2023.04.019>
- [23] T Martin-Noguerol, F Paulano-Godino, R López-Ortega, JM Górriz, RF Rascos, and A Luna. 2021. Artificial intelligence in radiology: relevance of collaborative work between radiologists and engineers for building a multidisciplinary team. *Clinical Radiology* 76, 5 (2021), 317–324. <https://doi.org/10.1016/j.crad.2020.11.113>
- [24] Claudia Mello-Thoms and Carlos AB Mello. 2023. Clinical applications of artificial intelligence in radiology. *The British Journal of Radiology* 96 (2023), 20221031. <https://doi.org/10.1259/bjr.20221031>
- [25] Seong K Mun, Shih-Chung B Lo, Kenneth Wong, Dow-Mu Koh, and Fred Prior. 2023. Emerging Value-Based Radiology in the Era of Artificial Intelligence. *Medical Research Archives* 11, 5 (2023). <https://doi.org/10.18103/mra.v11i5.3915>
- [26] European Society of Radiology (ESR) communications@myesr.org Becker Christoph D. Kotter Elmar Fournier Laure Marti-Bonmati Luis. 2022. Current practical experience with artificial intelligence in clinical radiology: a survey of the European Society of Radiology. *Insights into imaging* 13, 1 (2022), 107. <https://doi.org/10.1186/s13244-022-01247-y>
- [27] Bhavik N Patel, Louis Rosenberg, Gregg Willcox, David Baltaxe, Mimi Lyons, Jeremy Irvin, Pranav Rajpurkar, Timothy Amrhein, Rajan Gupta, Safwan Halabi, et al. 2019. Human–machine partnership with artificial intelligence for chest radiograph diagnosis. *NPJ digital medicine* 2, 1 (2019), 111. <https://doi.org/10.1038/s41746-019-0189-7>
- [28] Michael N Patlas, Douglas S Katz, Mariano Scaglione, et al. 2019. *Errors in emergency and trauma radiology*. Springer, Switzerland. <https://doi.org/10.1007/978-3-030-05548-6>
- [29] Eric W Prince, Todd C Hankinson, and Carsten Görg. 2023. The Iterative Design Process of an Explainable AI Application for Non-Invasive Diagnosis of CNS Tumors: A User-Centered Approach. In *2023 Workshop on Visual Analytics in Healthcare (VAHC)*. IEEE, IEEE Computer Society, Los Alamitos, CA, USA, 7–13. <https://doi.org/10.1109/VAHC60858.2023.00008>
- [30] Michael P Recht, Marc Dewey, Keith Dreyer, Curtis Langlotz, Wiro Niessen, Barbara Prainsack, and John J Smith. 2020. Integrating artificial intelligence into the clinical practice of radiology: challenges and recommendations. *European radiology* 30 (2020), 3576–3584. <https://doi.org/10.1007/s00330-020-06672-5>
- [31] Damian Scheek, Mohammad H Rezade Mehrizi, and Erik Ranschaert. 2021. Radiologists in the loop: the roles of radiologists in the development of AI applications. *European Radiology* 31 (2021), 7960–7968. <https://doi.org/10.1007/s00330-021-07879-w>
- [32] Rishi P Singh, Grant L Hom, Michael D Abramoff, J Peter Campbell, Michael F Chiang, et al. 2020. Current challenges and barriers to real-world artificial intelligence adoption for the healthcare system, provider, and the patient. *Translational Vision Science & Technology* 9, 2 (2020), 45–45. <https://doi.org/10.1167/tvst.9.2.45>
- [33] Lea Strohm, Charisma Hehakaya, Erik R Ranschaert, Wouter PC Boon, and Ellen HM Moors. 2020. Implementation of artificial intelligence (AI) applications in radiology: hindering and facilitating factors. *European radiology* 30 (2020), 5525–5532. <https://doi.org/10.1007/s00330-020-06946-y>
- [34] Kicky G van Leeuwen, Maarten de Rooij, Steven Schalekamp, Bram van Ginneken, and Matthieu JCM Rutten. 2024. Clinical use of artificial intelligence products for radiology in the Netherlands between 2020 and 2022. *European Radiology* 34 (2024), 348–354. <https://doi.org/10.1007/s00330-023-09991-5>
- [35] Oleksandra Vereschak, Gilles Bailly, and Baptiste Caramiaux. 2021. How to evaluate trust in AI-assisted decision making? A survey of empirical methodologies. *Proceedings of the ACM on Human-Computer Interaction* 5, CSCW2 (2021), 1–39. <https://doi.org/10.1145/3476068>
- [36] Julian L Wichmann, Martin J Willeminck, and Carlo N De Cecco. 2020. Artificial intelligence and machine learning in radiology: current state and considerations for routine clinical implementation. *Investigative Radiology* 55, 9 (2020), 619–627. <https://doi.org/10.1097/RLI.0000000000000673>
- [37] Ling Yang, Ioana Cezara Ene, Reza Arabi Belaghi, David Koff, Nina Stein, and Pasqualina Santaguida. 2022. Stakeholders' perspectives on the future of artificial intelligence in radiology: a scoping review. *European Radiology* 32, 3 (2022), 1477–1495. <https://doi.org/10.1007/s00330-021-08214-z>